

Michael A.C.M. Schmutzer

Dr. sc. nat.

Research experience

- 2025-2027 **Postdoc researcher**
Department of Earth Sciences, University of Oxford
Investigating the extinction of the ammonoids at the K-Pg boundary. Guest researcher supported by SNSF postdoc mobility grant P500PB_230716 / 1.
Principal Investigator: Prof. Erin Saupe
- 10-12/2024 **Postdoc researcher**
Department of Palaeontology, University of Zurich
Designing 3D model of Ammonite septa formation, and estimating respiration rates in trilobites.
Principal Investigator: Prof. Christian Klug
- 2017-2023 **PhD researcher**
Department of Evolutionary Biology and Environmental Studies, University of Zurich
Three research projects on the effects of randomness in molecular events on cell behaviour, and their evolutionary consequences.
Thesis: The role of Brownian-motion induced phenotypic stochasticity in evolution
Principal Investigator: Prof. Andreas Wagner
- 10-12/2016 **Research assistant**
Imperial College London
Predicting biodiversity in fragmented landscapes. Contributed to python package.
Principal Investigator: Prof. Robert Ewers
- 04-08/2016 **MSc research project**
Imperial College London
Building a model of bacterial speciation in the face of continuous gene flow.
Thesis: Modelling the evolution of gene-specific ecological divergence in bacteria
Principal Investigator: Prof. Timothy Barraclough
- 2015 **BSc research project**
Imperial College London
Analysing an ethnomusicology dataset using evolutionary methods.
Thesis: Stepping towards a geography of sound
Principal Investigator: Prof. Armand Leroi
- 2014 **Summer internship**
Natural History Museum London
Investigating PCR methods for amplifying degraded DNA from museum specimen.
Principal Investigator: Prof. Alfried Vogler

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Scientific communication

Scientific publications

- 2024 **M Schmutzer**, P Dasmeh, A Wagner. Frustration can limit the adaptation of promiscuous enzymes through gene duplication and specialisation. *Journal of Molecular Evolution*. 92 (2) 104-120
- 2023 **M Schmutzer**, A Wagner. Not quite lost in translation: Mistranslation alters adaptive landscape topography and the dynamics of evolution. *Molecular Biology and Evolution*. 40 (6) msad136
- 2020 **M Schmutzer**, A Wagner. Gene expression noise can promote the fixation of beneficial mutations in fluctuating environments. *PLOS Computational Biology*. 16 (10) e1007727
- 2019 **M Schmutzer**, TG Barraclough. The role of recombination, niche-specific gene pools and flexible genomes in the ecological speciation of bacteria. *Ecology and Evolution*. 9 (8) 4544-4556

Conference participation

- 2026 **European Geosciences Union General Assembly**
Poster session
- 2025 **Palaeontological Association Annual Meeting**
Poster session
- 2021 **Basel Computational Biology Conference**
Poster session – won best poster prize
- 2019 **Modelling in Ecology & Evolution Zurich**
Poster session

Education

- 2017-2023 **PhD Evolutionary Biology**
University of Zurich
- 2015-2016 **MSc Computational Methods in Ecology and Evolution**
Imperial College London
- 2012-2015 **BSc Biology**
Imperial College London

Won UZH PhD Forschungskredit grant No. FK-21-103

PhD courses

- Causal Inference for Biologists Graduate Workshop
- Doing Creative Research
- Teaching Science at University
- Zürich Interaction Seminar
- Project Design for Early Stage Researchers
- Scientific Writing for Evolutionary Biologists
- Filmmaking for Scientists

BSc Biology details

- 3rd year: Population and community ecology, evolutionary biology, biodiversity genomics; *Convenor's prize for biodiversity genomics*
- 2nd year: Cell development & behaviour, behavioural ecology, parasitology
- Tutored dissertation: 4000 word literature review ("Innovation by error? How novel genes shape adaptation")
- First year biology prize

BSc Biology Imperial Horizons

- Origins of Modern Science, 1550-1750
- Philosophy and the Sciences
- Philosophy: The Pragmatic Horizon
- Philosophy: Science and Beauty

Teaching & Supervision

2021	UZH BIO 352 Principles of Evolution: Practice <i>Supervising bachelor student project</i>
2019-2022	UZH BIO 334 Practical Bioinformatics <i>Teaching assistant</i>
2018-2022	UZH BIO 351 Principles of Evolution <i>Teaching assistant</i>
2018	UZH BIO 134 Programming in Biology <i>Teaching assistant</i>
2016	ICL MSc Computations Methods in Ecology and Evolution <i>Teaching assistant</i>

Developed self-paced online tools for BIO 334 and BIO 351 during the COVID-lockdowns in 2020.

Research & other skills

Research skills

- Scientific model construction as a vehicle for hypothesis testing; validation and analysis
- Parametric & non-parametric statistics, general linear models, non-linear least squares regression, cluster analysis, multivariate statistics, network analysis, linear optimization, agent-based modelling

- Fundamentals of causal inference
- Collaborative and independent project planning and execution

IT skills

- Proficient in python, R, bash, constructing, profiling, debugging & documenting python packages
- Familiar with scipy, numpy, matplotlib, multiprocessing, unit-testing, pandas
- Version control (git), L^AT_EX, SQLite
- Cluster computing & embarrassingly parallel simulations
- Fundamentals of flux balance analysis
- Palaeo-GIS processing in R

Communication & Languages

- Experienced presenter
- Trained & experienced in scientific writing
- Trained in basics of pedagogy
- Trained in basics of film-making
- *Proficient:* English, German, Dutch
Intermediate: French

Extracurricular interests

2019-2024	Filmstelle VSETH <i>Head of programming (2020-2021, finding & negotiating film rights, leading discussions), projection (DCP, 35mm), event organization</i>
2013-2015	Filmsoc Imperial College London <i>Vice-president, President</i>
ONGOING	Hobbies <i>Egyptian hieroglyphs and grammar, origami, avid reader, especially on history</i>
	Oxford Gamelan Society <i>Learning to play various instruments in a traditional Indonesian gamelan ensemble</i>